

IntelliMeet — Full-Stack Setup & Deployment Master Guide

Complete Walkthrough: Backend Server, MongoDB Database, API Connectivity, Live Deployment (Vercel & Render) & Troubleshooting

Project: IntelliMeet (MERN + WebRTC + AI)	Live Frontend: https://intelli-meet-nine.vercel.app
Repository: github.com/shreyay1310-svg/intelliMeet	Render Deploy Blueprint: 1-Click via render.yaml

1. System Architecture & Real-Time Connectivity

IntelliMeet is an enterprise meeting and collaboration suite built with a modern MERN stack. The application is bifurcated into an independent frontend single-page application (React 18 + Vite) and a scalable, event-driven Node.js backend with persistent WebSocket signaling for WebRTC peer connections.

Layer	Technology	Deployment Host	Role & Connectivity
Frontend	React 18, Vite, TypeScript, Tailwind	Vercel (Edge CDN)	Serves UI, connects via HTTP REST to <code>/api</code> and Socket.io for WebRTC signaling.
Backend	Node.js, Express, TypeScript, Socket.io	Render.com (Web Service)	REST endpoints, JWT verification, real-time room signaling, and AI summarization engine.
Database	MongoDB Atlas Cloud / Local 7.0	MongoDB Cloud (M0 Free)	NoSQL document storage for Users, Meetings, Action Items, Transcripts, and Audit Logs.

2. Database Configuration: MongoDB vs SQL Server (SSMS)

Should you switch the database to SQL Server (SSMS)?

Answer: NO. Migrating to SQL Server is strongly discouraged for this project due to the following technical constraints:

- Mongoose ODM Architecture:** All models (*User.ts*, *Meeting.ts*, *Task.ts*, *Summary.ts*, *Transcript.ts*) use Mongoose schemas. Mongoose cannot communicate with Microsoft SQL Server; moving would require rewriting the entire backend in TypeORM/Prisma with custom relational SQL queries.
- Hierarchical Data Models:** Meeting rooms store dynamic participant arrays, live chat streams, and AI action bullets. In MongoDB, these are cleanly persisted as nested JSON subdocuments. In SQL Server, it would require 8+ relational tables and expensive multi-table JOINS.
- Cloud Hosting & Cost:** MongoDB Atlas provides a 100% free lifetime cloud tier (M0) that seamlessly links with Render and Vercel. Hosting a dedicated MS SQL Server instance on the cloud is expensive and non-trivial on free-tier platforms.
- Visual Management Alternative:** If you desire a graphical studio interface identical to SQL Server Management Studio (SSMS), install **MongoDB Compass**. It provides visual schemas, live document editors, and index inspection.

3. Live Production Deployment: Vercel + Render

Because Socket.io requires a persistent TCP connection for WebRTC signaling, Vercel Serverless functions cannot host the backend. The optimal production architecture is: **Vercel for Frontend** and **Render for Backend**.

Part A: Frontend Deployment on Vercel

- Active Production Link:** <https://intelli-meet-nine.vercel.app>
- Status:** Deployed and auto-synced with the GitHub repository [shreyay1310-svg/intelliMeet](https://github.com/shreyay1310-svg/intelliMeet).
- Vercel Settings:** Root Directory = `client` | Framework = `Vite` | Output = `dist`.

Part B: 1-Click Backend Deployment on Render (Automated via Blueprint)

We configured and pushed an automated `render.yaml` Blueprint to your repository. You do not need to configure build scripts manually:

- Open the Direct Deploy Link: **`https://render.com/deploy?repo=https://github.com/shreyay1310-svg/intelliMeet`**
- Sign in with GitHub and authorize repository access.
- Render automatically parses `render.yaml`, setting Service = `intellimeet-api`, Root = `server`, Build = `npm install && npm run build`, and Start = `npm run start`.
- Click **'Apply'**. Render will provision the container and provide your live backend URL (e.g., `https://intellimeet-api.onrender.com`).

5. Go to Vercel -> Project Settings -> **Environment Variables** -> Add VITE_API_URL = <https://intellimeet-api.onrender.com> and click **Redeploy**.

4. Diagnostic & Troubleshooting Log (Issues Solved Today)

During setup, four distinct runtime and environment blockers were systematically diagnosed and resolved:

Symptom / Error	Root Cause Identified	Resolution Applied
Error: Unexpected end of JSON input	Frontend called <code>/api/auth/register</code> on Vercel where no backend was running, returning an empty HTML 404.	Refactored <code>client/src/services/api.ts</code> to dynamically read <code>VITE_API_URL</code> with robust non-JSON fallback and error reporting.
Error: Request failed with status 500	MongoDB Atlas authentication failure: bad auth (code: 8000) caused by angle brackets <code>< ></code> around password in <code>MONGO_URI</code> .	Sanitized connection string format in <code>server/.env</code> . Guided Atlas password reset and IP whitelist (0.0.0.0/0).
fatal: index.lock exists & refsPEC main does not match	PowerShell session was positioned in user home folder <code>C:\Users\shreya</code> rather than workspace root.	Navigated to <code>C:\Users\shreya\OneDrive\Desktop\intelliMeet</code> , added Git to PATH, committed, and successfully pushed to origin.
npm error ENOSPC: no space left on device	System C: drive was at 100% capacity (0 MB free space), causing child process crashes (EPERM) and build failures.	Cleaned 1.67 GB of cached temporary build files to restore operating system stability and allow compilation.

5. Local Development Execution Guide

To run the full stack simultaneously on your local computer:

Action	Terminal Command
Step 1: Start Backend (Port 5000)	<code>cd server && npm run dev</code>
Step 2: Start Frontend (Port 5173)	<code>cd client && npm run dev</code>
Or Start Both Together from Root	<code>npm run dev</code>

6. Pre-Seeded Workspace Credentials

The platform automatically provisions verified seed credentials for instant evaluation:

Role	Email	Password	Designation & Team
Employee	shreya@zidio.in	Password123!	Product Designer, Product Team
Administrator	admin@zidio.in	Password123!	Enterprise Admin, Executive Team

IntelliMeet Documentation & Installation Blueprint | Generated automatically for Shreya Yadav | Repository: <https://github.com/shreyay1310-svg/intelliMeet>